

Foundational AI Reading Package

A structured guide to the reconstructed 'Ilya Sutskever 30 papers' list

[Source collection: 30papers.com](https://30papers.com)

Scope note: This is an unofficial reconstruction of a widely discussed reading list, not an authenticated publication by Ilya Sutskever. This package contains links and study notes, not copies of the papers.

How to use this package

- First pass: read the introduction, figures, and conclusion of each item; write down the central mechanism in one sentence.
- Second pass: derive the main equations and track tensor dimensions or probabilistic factorizations.
- Third pass: reproduce one small experiment or implementation per section.
- Do not read strictly chronologically. Use the recommended order below to reach Transformers and scaling laws quickly.

Recommended fast-track order

1. CS231n: Convolutional Neural Networks for Visual Recognition
2. ImageNet Classification with Deep Convolutional Neural Networks
3. Deep Residual Learning for Image Recognition
4. Understanding LSTM Networks
5. Sequence to Sequence Learning with Neural Networks
6. Neural Machine Translation by Jointly Learning to Align and Translate
7. Attention Is All You Need
8. The Annotated Transformer
9. Scaling Laws for Neural Language Models
10. Neural Turing Machines
11. Neural Message Passing for Quantum Chemistry
12. Keeping Neural Networks Simple by Minimizing the Description Length of the Weights

- 13. A Tutorial Introduction to the Minimum Description Length Principle
- 14. The First Law of Complexodynamics
- 15. Machine Super Intelligence

Ten-week curriculum

Week 1: CS231n foundations; implement a small multilayer perceptron and convolution layer.

Week 2: AlexNet, ResNet, and identity mappings; reproduce a residual block.

Week 3: RNN intuition, LSTM, and recurrent regularization.

Week 4: Seq2seq, attention, and pointer networks.

Week 5: Transformer paper plus the annotated implementation.

Week 6: Neural Turing Machines and relational reasoning.

Week 7: Graph message passing and molecular modeling.

Week 8: Scaling laws and GPipe; reason about compute, data, and parallelism.

Week 9: MDL, Bayesian compression, and Kolmogorov complexity.

Week 10: Complexity, lossy representation, and formal views of intelligence.

Foundations / Vision

1. CS231n: Convolutional Neural Networks for Visual Recognition

Stanford course notes

Why it matters: Builds the practical vocabulary for convolution, receptive fields, optimization, and visual recognition.

Prerequisites: Linear algebra, multivariable calculus, basic probability

Reading focus: Focus on convolution geometry, backpropagation through layers, normalization, regularization, and optimization.

Open link: <https://cs231n.github.io/>

2. ImageNet Classification with Deep Convolutional Neural Networks

Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton (2012)

Why it matters: The AlexNet paper that triggered the modern deep-learning wave in computer vision.

Prerequisites: Neural-network basics, convolution

Reading focus: Separate the architectural contribution from the training-system contribution: ReLU, dropout, augmentation, GPUs, and scale.

Open link:

https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf

3. Deep Residual Learning for Image Recognition

Kaiming He et al. (2015)

Why it matters: Introduces residual blocks, making very deep networks substantially easier to optimize.

Prerequisites: CNNs, gradient descent

Reading focus: Understand why learning $F(x)=H(x)-x$ can be easier than learning $H(x)$ directly, and distinguish optimization degradation from overfitting.

Open link: <https://arxiv.org/pdf/1512.03385>

4. Multi-Scale Context Aggregation by Dilated Convolutions

Fisher Yu, Vladlen Koltun (2015)

Why it matters: Shows how dilated convolutions enlarge receptive fields without losing resolution.

Prerequisites: CNNs, convolution arithmetic

Reading focus: Track the receptive field layer by layer and compare dilation with pooling and stride.

Open link: <https://arxiv.org/pdf/1511.07122>

5. Identity Mappings in Deep Residual Networks

Kaiming He et al. (2016)

Why it matters: Analyzes why clean identity skip paths improve signal and gradient propagation in residual networks.

Prerequisites: ResNet paper, chain rule

Reading focus: Follow the algebra of forward and backward propagation through identity shortcuts; note the pre-activation design.

Open link: <https://arxiv.org/pdf/1603.05027>

Sequence Models

6. The Unreasonable Effectiveness of Recurrent Neural Networks

Andrej Karpathy (2015)

Why it matters: An intuitive, code-driven introduction to character-level RNNs and sequence generation.

Prerequisites: Basic neural networks, Python

Reading focus: Use it to build intuition for hidden state, teacher forcing, sampling temperature, and long-range dependence.

Open link: <https://karpathy.github.io/2015/05/21/rnn-effectiveness/>

7. Understanding LSTM Networks

Christopher Olah (2015)

Why it matters: A visual explanation of LSTM gates and the cell-state pathway.

Prerequisites: RNN basics

Reading focus: Translate every diagram into equations and identify where multiplicative gates control information flow.

Open link: <https://colah.github.io/posts/2015-08-Understanding-LSTMs/>

8. Recurrent Neural Network Regularization

Wojciech Zaremba, Ilya Sutskever, Oriol Vinyals (2014)

Why it matters: Establishes an effective way to apply dropout in stacked RNNs without disrupting recurrent dynamics.

Prerequisites: RNNs, dropout

Reading focus: Notice exactly which connections receive dropout and which do not; connect this choice to temporal memory.

Open link: <https://arxiv.org/pdf/1409.2329>

9. Deep Speech 2: End-to-End Speech Recognition in English and Mandarin

Dario Amodei et al. (2015)

Why it matters: Demonstrates a large-scale end-to-end speech system and the engineering required to train it efficiently.

Prerequisites: RNNs, CTC helpful

Reading focus: Read for system design: data, distributed training, recurrent architectures, decoding, and robustness.

Open link: <https://arxiv.org/pdf/1512.02595>

Sequence-to-Sequence and Attention

10. Sequence to Sequence Learning with Neural Networks

Ilya Sutskever, Oriol Vinyals, Quoc Le (2014)

Why it matters: Introduces the encoder-decoder LSTM formulation for general sequence transduction.

Prerequisites: LSTMs, conditional probability

Reading focus: Write the factorization of $p(y|x)$, understand the fixed-vector bottleneck, and study the source-sequence reversal trick.

Open link: <https://arxiv.org/pdf/1409.3215>

11. Order Matters: Sequence to Sequence for Sets

Oriol Vinyals, Samy Bengio, Manjunath Kudlur (2015)

Why it matters: Examines how sequence models can process sets despite the arbitrary ordering of set elements.

Prerequisites: Seq2seq, permutation invariance

Reading focus: Ask which parts of the model are order-sensitive and how training or search handles latent order.

Open link: <https://arxiv.org/pdf/1511.06391>

12. Neural Machine Translation by Jointly Learning to Align and Translate

Dzmitry Bahdanau, Kyunghyun Cho, Yoshua Bengio (2014)

Why it matters: Introduces soft attention for neural machine translation, removing the single-vector bottleneck.

Prerequisites: Seq2seq, softmax, differentiable optimization

Reading focus: Derive the attention weights and context vector; interpret attention as content-based addressing.

Open link: <https://arxiv.org/pdf/1409.0473>

13. Pointer Networks

Oriol Vinyals, Meire Fortunato, Navdeep Jaitly (2015)

Why it matters: Uses attention as an output distribution over input positions, enabling variable-size combinatorial outputs.

Prerequisites: Attention, seq2seq

Reading focus: Contrast a fixed vocabulary softmax with a pointer distribution over source indices.

Open link: <https://arxiv.org/pdf/1506.03134>

14. Attention Is All You Need

Ashish Vaswani et al. (2017)

Why it matters: Introduces the Transformer: self-attention, multi-head attention, positional encoding, and fully parallel training.

Prerequisites: Linear algebra, probability, seq2seq

Reading focus: Derive scaled dot-product attention, track tensor shapes, and understand the roles of residual paths, normalization, and masking.

Open link: <https://arxiv.org/pdf/1706.03762>

15. The Annotated Transformer

Harvard NLP

Why it matters: A line-by-line implementation and explanation of the original Transformer.

Prerequisites: Transformer paper, PyTorch familiarity

Reading focus: Use it after the paper to connect equations with code; verify masks, dimensions, and training schedule.

Open link: <https://nlp.seas.harvard.edu/annotated-transformer/>

Memory, Reasoning, and Graphs

16. Neural Turing Machines

Alex Graves, Greg Wayne, Ivo Danihelka (2014)

Why it matters: Augments neural networks with differentiable external memory and read/write heads.

Prerequisites: RNNs, attention, differentiable programming

Reading focus: Study content-based and location-based addressing and how the controller learns algorithms.

Open link: <https://arxiv.org/pdf/1410.5401>

17. A Simple Neural Network Module for Relational Reasoning

Adam Santoro et al. (2017)

Why it matters: Introduces Relation Networks, explicitly aggregating pairwise object relations.

Prerequisites: Basic deep learning

Reading focus: Write the core sum over object pairs and identify the inductive bias toward relational structure.

Open link: <https://arxiv.org/pdf/1706.01427>

18. Relational Recurrent Neural Networks

Adam Santoro et al. (2018)

Why it matters: Combines recurrent state with relational computation to model interactions among entities over time.

Prerequisites: RNNs, relation networks, attention

Reading focus: Understand how memory slots interact and how relational processing changes recurrent state updates.

Open link: <https://arxiv.org/pdf/1806.01822>

19. Neural Message Passing for Quantum Chemistry

Justin Gilmer et al. (2017)

Why it matters: Unifies many graph neural networks under a message-passing framework and applies them to molecular prediction.

Prerequisites: Graphs, neural networks

Reading focus: Separate message, aggregation, update, and readout functions; note invariance to node ordering.

Open link: <https://arxiv.org/pdf/1704.01212>

Scaling and Systems

20. Scaling Laws for Neural Language Models

Jared Kaplan et al. (2020)

Why it matters: Empirically characterizes power-law relationships among model size, data, compute, and language-model loss.

Prerequisites: Log-log plots, optimization, language modeling

Reading focus: Study the fitted exponents, the compute-allocation argument, and the distinction between undertraining and irreducible loss.

Open link: <https://arxiv.org/pdf/2001.08361>

21. GPipe: Efficient Training of Giant Neural Networks using Pipeline Parallelism

Yanping Huang et al. (2018)

Why it matters: Introduces pipeline parallelism with micro-batches for training models too large for one accelerator.

Prerequisites: Distributed systems, backpropagation

Reading focus: Track pipeline bubbles, partitioning, activation storage, and the trade-off between throughput and memory.

Open link: <https://arxiv.org/pdf/1811.06965>

Information Theory and Compression

22. Keeping Neural Networks Simple by Minimizing the Description Length of the Weights

Geoffrey Hinton, Drew van Camp (1993)

Why it matters: Connects Bayesian learning, compression, and regularization through a description-length objective.

Prerequisites: Probability, KL divergence, Bayesian inference

Reading focus: Relate the variational distribution over weights to the coding cost and generalization pressure.

Open link: <https://www.cs.toronto.edu/~hinton/absps/colt93.pdf>

23. A Tutorial Introduction to the Minimum Description Length Principle

Peter Grünwald (2004)

Why it matters: Explains model selection as finding the shortest total code for model plus data.

Prerequisites: Probability, information theory

Reading focus: Distinguish two-part codes, universal codes, stochastic complexity, and the connection to Occam's razor.

Open link: <https://arxiv.org/pdf/math/0406077>

24. Kolmogorov Complexity

Chapter/topic commonly read via information-theory texts

Why it matters: Provides the algorithmic notion of description length underlying universal induction and MDL.

Prerequisites: Computability, information theory

Reading focus: Focus on invariance up to an additive constant, incomputability, and the difference from Shannon entropy.

Open link: https://www.scholarpedia.org/article/Algorithmic_information_theory

Complexity and Intelligence

25. The First Law of Complexodynamics

Scott Aaronson (essay/lecture)

Why it matters: Explores why macroscopic complexity can rise temporarily even while entropy increases.

Prerequisites: Statistical mechanics, entropy

Reading focus: Separate thermodynamic entropy from intuitive structural complexity and examine the proposed rise-then-fall behavior.

Open link: <https://www.scottaaronson.com/papers/complex.pdf>

26. Quantifying the Rise and Fall of Complexity in Closed Systems: The Coffee Automaton

Scott Aaronson, Sean Carroll, Lauren Ouellette (2014)

Why it matters: Studies candidate complexity measures in simple closed systems that mix from ordered to equilibrium states.

Prerequisites: Probability, cellular automata, statistical mechanics

Reading focus: Compare entropy with measures intended to peak at intermediate structure; inspect where each measure succeeds or fails.

Open link: <https://arxiv.org/pdf/1405.6903>

27. Variational Lossy Autoencoder

Xi Chen et al. (2016)

Why it matters: Separates representation capacity from reconstruction detail using a variational lossy autoencoding objective.

Prerequisites: VAEs, variational inference, information theory

Reading focus: Understand the role of the autoregressive decoder and why the latent code is pressured to capture global structure.

Open link: <https://arxiv.org/pdf/1611.02731>

28. Machine Super Intelligence

Shane Legg (PhD thesis, 2008)

Why it matters: A broad treatment of machine intelligence, universal intelligence measures, and possible superintelligence.

Prerequisites: Computability, probability, reinforcement learning

Reading focus: Read selectively: definitions of intelligence, universal agents, measurement proposals, and limitations of formalization.

Open link: https://www.vetta.org/documents/Machine_Super_Intelligence.pdf

Reading checklist

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1	CS231n: Convolutional Neural Networks for Visual Recognition	☐	☐
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4	Multi-Scale Context Aggregation by Dilated Convolutions	☐	☐
5	Identity Mappings in Deep Residual Networks	☐	☐
6	The Unreasonable Effectiveness of Recurrent Neural Networks	☐	☐
7	Understanding LSTM Networks	☐	☐
8	Recurrent Neural Network Regularization	☐	☐
9	Deep Speech 2: End-to-End Speech Recognition in English and Mandarin	☐	☐
10	Sequence to Sequence Learning with Neural Networks	☐	☐
11	Order Matters: Sequence to Sequence for Sets	☐	☐
12	Neural Machine Translation by Jointly Learning to Align and Translate	☐	☐
13	Pointer Networks	☐	☐
14	Attention Is All You Need	☐	☐
15	The Annotated Transformer	☐	☐
16	Neural Turing	☐	☐

	Machines		
17	A Simple Neural Network Module for Relational Reasoning	☐	☐
18	Relational Recurrent Neural Networks	☐	☐
19	Neural Message Passing for Quantum Chemistry	☐	☐
20	Scaling Laws for Neural Language Models	☐	☐
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23	A Tutorial Introduction to the Minimum Description Length Principle	☐	☐
24	Kolmogorov Complexity	☐	☐
25	The First Law of Complexodynamics	☐	☐
26	Quantifying the Rise and Fall of Complexity in Closed Systems: The Coffee Automaton	☐	☐
27	Variational Lossy Autoencoder	☐	☐
28	Machine Super Intelligence	☐	☐

Availability and copyright

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